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$$f_n(t)y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

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Definition (Initial value problems, IVPs)

$$\text{Usually } y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots \text{ and } y^{(n-1)}(t_0) = y_0^{(n-1)}$$

Lecture 2

- In Lecture 1, we defined linear ODEs+IVPs and revised integrating factors
- In Lecture 2, we will:
  1. Form a connection between 1st-order ODEs and linear algebra
  2. Express ODE solutions in terms of complementary and particular solutions
  3. Extend concepts to 2nd-order ODEs
  4. Learn how to use the reduction of order method

Connection to linear algebra

- Linear algebra: $\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$ where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{w} \in \mathbb{R}^m$.

*In Topic 4 we will extend the idea of these finite linear combinations to an infinite series. If you enjoy this aspect of the course then take MAST30026 Metric & Hilbert Spaces to gain a more detailed understanding.

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- $\mathbf{A} : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation:

$$\mathbf{A}(\boldsymbol{u} + \boldsymbol{v}) = \mathbf{A}\boldsymbol{u} + \mathbf{A}\boldsymbol{v} \quad \text{and} \quad \mathbf{A}(k\boldsymbol{u}) = k\mathbf{A}\boldsymbol{u} \quad \text{where} \quad \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n;\ k \in \mathbb{R}$$

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- $\frac{\mathrm{d}}{\mathrm{d}t}$ and $f(t)\cdot$ are also linear transformations but they operate on functions.*
- $\mathsf{T} = \frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$ is a *linear transformation*

$$(\mathsf{T}y)(t) = \frac{\mathrm{d}y}{\mathrm{d}t} + f(t)y(t)$$

...you should check the definition is satisfied, i.e.,

$$\mathsf{T}(x+y) = \mathsf{T}x + \mathsf{T}y \quad \text{and} \quad \mathsf{T}(k\,y) = k\,\mathsf{T}y$$

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We have the following connections

$\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$	$\leftrightarrow$	$y'(t) + f(t)y(t) = g(t)$
$\boldsymbol{v}$	$\leftrightarrow$	$y(t)$
$\boldsymbol{w}$	$\leftrightarrow$	$g(t)$
$\mathbf{A}$	$\leftrightarrow$	$\frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$

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- General solution: $\boldsymbol{v} = \boldsymbol{v}_c + \boldsymbol{v}_p$.

The same procedure works for linear ODEs

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- General solution: $y = y_c + y_p$

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We considered the linear, first-order, non-homogeneous DE, using an IF.

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Second order linear ODEs

- Using notation from the linear algebra analogy, consider ODEs of form

$$(\mathsf{T}y)(t) = g(t) \quad \text{where} \quad \mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

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Topic 6: Fourier transforms

Trial solutions for constant coefficient ODEs (revision)

Consider $\mathsf{T}y = 0$, where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}}{\mathrm{d}t} + c \quad \text{such that} \quad b, c \in \mathbb{R}$$

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- Then $\lambda_\pm = \dfrac{-b \pm \sqrt{b^2 - 4c}}{2}$
- Therefore, $y = C_+\mathrm{e}^{\lambda_+ t} + C_-\mathrm{e}^{\lambda_- t}$ if $b^2 - 4c \neq 0$ so that $\lambda_+ \neq \lambda_-$

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Reduction of order—general theory

Definition

Let $v(t)$ be a non-zero solution of $\mathsf{T}v = 0$ where T is an arbitrary second-order linear operator, then the general solution to $\mathsf{T}y = g$ can be found in the form $y(t) = u(t)v(t)$.

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- $\mathsf{T}v = 0$ and hence $\mathsf{T}y = u''v + 2u'v' + f_1u'v = g$. <u>No f_0 (explicitly)</u>

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Reduction of order—general theory

Definition

Let $v(t)$ be a non-zero solution of $\mathsf{T}v = 0$ where T is an arbitrary second-order linear operator, then the general solution to $\mathsf{T}y = g$ can be found in the form $y(t) = u(t)v(t)$.

- Chain rule gives $y' = u'v + uv'$ and $y'' = u''v + 2u'v' + uv''$
- Therefore

$$\begin{aligned} \mathsf{T}y &= u''v + 2u'v' + uv'' + f_1(u'v + uv') + f_0uv \\ &= u\,\mathsf{T}v + u''v + 2u'v' + f_1u'v \end{aligned}$$

- $\mathsf{T}v = 0$ and hence $\mathsf{T}y = u''v + 2u'v' + f_1u'v = g$. <u>No f_0 (explicitly)</u>
- Rearranging gives $u'' + \left(\frac{2v'}{v} + f_1\right)u' = \frac{g}{v}$. <u>What do you notice?</u>

- Let $w = u'$, so $w' + \left(\frac{2v'}{v} + f_1\right) w = \frac{g}{v}$

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Note: the two integrals will give us the required two unknown constants.

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Finding the first solution for $\mathsf{T}v = 0$ is usually the difficult task.

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Therefore

$$y = t\,u = -\frac{C_1}{2t} + C_2 t$$

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$$y = t\left(\frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2} - \frac{C_1}{2t^2} + C_2 \right) = \left(1 - \frac{1}{t}\right)\mathrm{e}^t - \frac{C_1}{2t} + C_2 t$$

Workings

$$
\begin{aligned}
\int \left\{ \frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3} \right\} \mathrm{d}t &= \int \frac{\mathrm{e}^t}{t} \mathrm{d}t - 2\left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) + 2\left(-\frac{\mathrm{e}^t}{2t^2} + \int \frac{\mathrm{e}^t}{2t^2} \mathrm{d}t \right) \\
&= -\frac{\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t} - \int \frac{\mathrm{e}^t}{t} \mathrm{d}t + \left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) \\
&= \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2}
\end{aligned}
$$